

1973-2021

CLIMATE CRISIS

The idea that agricultural practices such as farming, irrigation, and forest clearance could affect local patterns of temperature and rainfall dates back to the ancient world. However, the notion that global climate can be affected by human activity emerged only in the 19th century. Scientists including Joseph Fourier and John Tyndall realized that certain gases in Earth's atmosphere – notably carbon dioxide and methane – absorbed heat from the Sun and warmed the atmosphere, a phenomenon that became known as the “greenhouse effect”. By the mid-20th century, research had established a link between increasing amounts of carbon dioxide in the atmosphere – the result of

burning fossil fuels – and rising global temperatures. These findings led US oceanographer Roger Revelle to write: “Human beings are now carrying out a large scale geophysical experiment...”.

Atmospheric monitoring by the US geochemist Charles Keeling in the 1950s showed a clear pattern of rising carbon dioxide concentrations, and the new science of computer modelling used such data to predict future climate scenarios; these included higher global temperatures, droughts, melting glaciers and ice caps, and rising sea levels. In 1965, climate change appeared on the political agenda as a US Presidential Advisory Committee called the greenhouse effect a matter of “real concern”.



KEY MOMENTS

1973 UN Conference on the Environment

Stockholm in Sweden hosted the world's first conference to address global environmental issues. Commitments were made to study the links between growth and environmental damage such as deforestation (*left*).

1989 IPCC assessments

The first report of the Intergovernmental Panel on Climate Change (IPCC) showed that temperatures had risen by 0.3–0.6°C (0.5–1.1°F) over the last 100 years. Its fifth report stated with 95 per cent certainty that humans were the “dominant cause” of global warming and climate change (*left*) since the 1950s.

1997-2021 Agreements and disagreements

International treaties such as the Kyoto Protocol (1997) and the 2015 Paris Agreement (*left*) sought commitment to reduce the output of greenhouse gases. Many countries failed to meet their targets; some, such as the US, withdrew for political reasons.

